

Multinomial Data

The multinomial distribution is an extension of the binomial where the response can take more than two values. Let Y_i be a random variable that falls into one of a finite number of categories, labeled $1, 2, \dots, J$. Let $p_{ij} = P(Y_i = j)$ so $\sum_{j=1}^J p_{ij} = 1$. As with binary data (the case where $J = 2$), we may encounter both grouped and ungrouped data. Let Y_{ij} be the number of observations falling into category j for group or individual i and let $n_i = \sum_j Y_{ij}$. For ungrouped data, $n_i = 1$ and one and only one of $Y_{i1}, \dots, Y_{iJ}$ is equal to one and the rest are zero. The Y_{ij} , conditional on the total n_i , follow a *multinomial* distribution:

$$P(Y_{i1} = y_{i1}, \dots, Y_{iJ} = y_{iJ}) = \frac{n_i!}{y_{i1}! \dots y_{iJ}!} p_{i1}^{y_{i1}} \dots p_{iJ}^{y_{iJ}}$$

We must also distinguish between *nominal* multinomial data where there is no natural order to the categories and *ordinal* multinomial data where there is an order. The *multinomial logit* model is intended for nominal data. It can be used for ordinal data, but the information about order will not be used.

7.1 Multinomial Logit Model

As with the binary response model, we must find a way to link the probabilities p_{ij} to the predictors x_i , while ensuring that the probabilities are restricted between zero and one. We can use a similar idea:

$$\eta_{ij} = x_i^T \beta_j = \log \frac{p_{ij}}{p_{i1}}, \quad j = 2, \dots, J$$

We must obey the constraint that $\sum_{j=1}^J p_{ij} = 1$, so it is convenient to declare one of the categories as the *baseline*, say, $j = 1$. So we get $p_{i1} = 1 - \sum_{j=2}^J p_{ij}$ and have:

$$p_{ij} = \frac{\exp(\eta_{ij})}{1 + \sum_{j=2}^J \exp(\eta_{ij})}$$

Note that $\eta_{i1} = 0$. It does not matter which category is declared as the baseline although some choices may be more convenient for interpretation. We may estimate the parameters of this model using maximum likelihood and then use the standard methods of inference.

Consider an example drawn from a subset of the 1996 American National Election Study (Rosenstone et al. (1997)). For simplicity, we consider only the age,

education level and income group of the respondents. Our response will be the party identification of the respondent: Democrat, Independent or Republican. The original data involved more than three categories; we collapse this to three, again for simplicity of the presentation.

```
data(nes96, package="faraway")
party <- nes96$PID
levels(party) <- c("Democrat", "Democrat", "Independent", "Independent",
  ↪ "Independent", "Republican", "Republican")
inca <- c(1.5, 4, 6, 8, 9.5, 10.5, 11.5, 12.5, 13.5, 14.5, 16, 18.5, 21, 23.5,
  ↪ 27.5, 32.5, 37.5, 42.5, 47.5, 55, 67.5, 82.5, 97.5, 115)
income <- inca[unclass(nes96$income)]
rnes96 <- data.frame(party, income, education=nes96$educ, age=nes96$
  ↪ age)
summary(rnes96)
```

	party	income	education	age
Democrat	:380	Min. : 1.5	MS : 13	Min. :19
Independent	:239	1st Qu.: 23.5	HSdrop: 52	1st Qu.:34
Republican	:325	Median : 37.5	HS :248	Median :44
		Mean : 46.6	Coll :187	Mean :47
		3rd Qu.: 67.5	CCdeg : 90	3rd Qu.:58
		Max. :115.0	BAdeg :227	Max. :91
			MAdeg :127	

The income variable in the original data was an ordered factor with income ranges. We have converted this to a numeric variable by taking the midpoint of each range. This corresponds to a score-based approach of handling interval-valued variables.

Let's start with a graphical look at the relationship between the predictors and the response. The response is at the individual level and so we need to group the data just to get a sense of how the party identification is associated with the predictors. Some manipulation of the data is facilitated with the `dplyr` package. We group the data by education and party affiliation, count the number in each category and then count the number in each education category, using this to compute the proportion supporting each party for each party affiliation. The resulting plot is shown in the first panel of Figure 7.1. We see that proportion of Democrats falls with educational status, reaching a plateau for the college educated. We see the proportion of Republicans rising with educational level and reaching a similar plateau.

```
library(dplyr)
egp <- group_by(rnes96, education, party) %>% summarise(count=n()) %>%
  ↪ group_by(education) %>% mutate(etotal=sum(count), proportion=
  ↪ count/etotal)
ggplot(egp, aes(x=education, y=proportion, group=party, linetype=party
  ↪ ))+geom_line()
```

A similar calculation is made for income but unfortunately there are relatively low counts in some income categories making the computation of a stable proportion difficult. To overcome this, we first group the income into seven groups of roughly equal size and then follow much the same calculation as for education.

```
igp <- mutate(rnes96, incomegp=cut_number(income,7)) %>% group_by(
  ↪ incomegp, party) %>% summarise(count=n()) %>% group_by(incomegp
  ↪ ) %>% mutate(etotal=sum(count), proportion=count/etotal)
ggplot(igp, aes(x=incomegp, y=proportion, group=party, linetype=party)
  ↪ )+geom_line()
```

The plot is shown in the second panel of Figure 7.1. As income increases, we observe

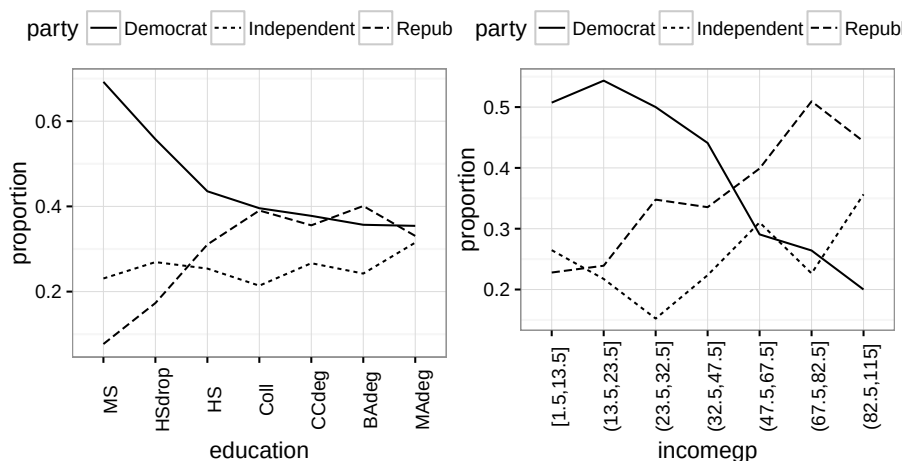


Figure 7.1 Relationship between party affiliation and education, age and income. Democrats are shown with solid line, Republicans with a dashed line and Independents with a dotted line. Education is categorized into seven levels described in the text. Income is in thousands of dollars.

an increase in the proportion of Republicans and Independents and a decrease in the proportion of Democrats. A similar plot can be made for age where the relationship of party to age is not clear. This is cross-sectional rather than longitudinal data, so we cannot say anything about what might happen to an individual with, for example, increasing income. We can only expect to make conclusions about the relative probability of party affiliations for different individuals with different incomes.

We might ask whether the trends we see in the observed proportions are statistically significant. We need to model the data to answer this question. We fit a multinomial logit model. The `multinom` function is part of the `nnet` package described in Venables and Ripley (2002):

```
library(nnet)
mmmod <- multinom(party ~ age + education + income, rnes96)
# weights: 30 (18 variable)
initial value 1037.090001
iter 10 value 990.568608
iter 20 value 984.319052
final value 984.166272
converged
```

The program uses the optimization method from the neural net trainer in `nnet` as described in Chapter 17 to compute the maximum likelihood. There is no deeper connection to neural networks — we just need the optimization found in that package.

We can select which variables to include in the model based the AIC criterion using a stepwise search method (output edited to show only the decision information):

```
mmodi <- step(mmod)
Start: AIC=2004.3
party ~ age + education + income
```

```

      Df    AIC
- education 6 1996.5
- age       16 2003.6
<none>     18 2004.3
- income    16 2045.9

```

```

Step: AIC=1996.5
party ~ age + income
      Df    AIC
- age    4 1993.4
<none>   6 1996.5
- income 4 2048.9

```

```

Step: AIC=1993.4
party ~ income
      Df    AIC
<none>  4 1993.4
- income 2 2045.3

```

At the first stage of the search, we see that omitting education would be the best option to reduce the AIC criterion. At the next step, age is removed resulting in a model with only income.

We can also use the standard likelihood methods to derive a test to compare nested models. For example, we can fit a model without education and then compare the deviances:

```

mmode <- multinom(party ~ age + income, rnes96)
deviance(mmode) - deviance(mmod)
[1] 16.206
pchisq(16.206, mmod$edf - mmode$edf, lower=F)
[1] 0.18198

```

We see that education is not significant relative to the full model. This may seem somewhat surprising given the plot in Figure 7.1, but the large differences between proportions of Democrats and Republicans occur for groups with low education which represent only a small number of people.

We can obtain predicted values for specified values of income. We can compute the probability of party affiliation for a range of incomes in $[0, 110]$.

```

inclevels <- 0:110
preds <- data.frame(income=inclevels, predict(mmodi, data.frame(income=
  ↪ inclevels), type="probs"))
library(tidyr)
lpred <- gather(preds, party, probability, -income)
ggplot(lpred, aes(x=income, y=probability, group=party, linetype=party)) +
  ↪ geom_line()

```

We see in Figure 7.2 how the probability of being Republican or Independent increases with income while the probability of being Democrat drops. The default form just gives the most probable category:

```

predict(mmodi, data.frame(income=inclevels))
[1] Democrat Democrat Democrat Democrat Democrat
....
[111] Republican

```

The multinomial logit model can also be used in a predictive sense to classify new observations. We can see how well the selected model does in predicting the party of the known individuals in our dataset:

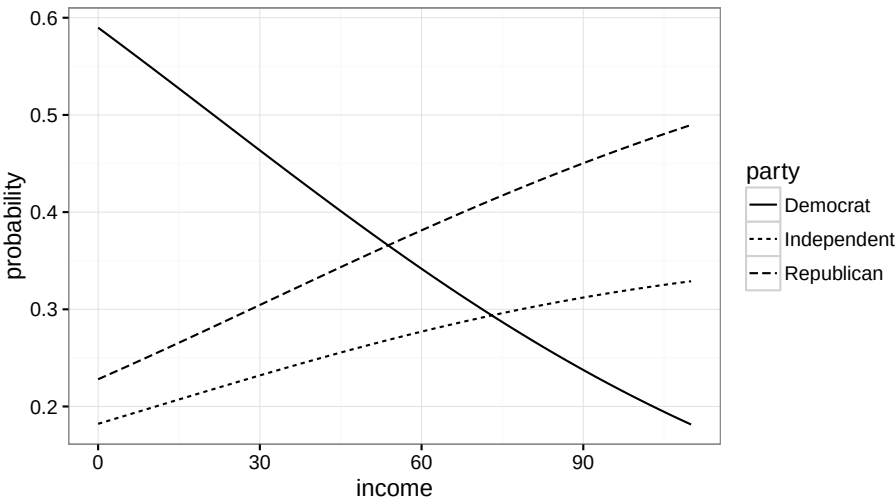


Figure 7.2 Predicted probabilities of party affiliation as income varies (thousands).

```
xtabs( ~ predict(mmodi) + rnes96$party)
```

	Democrat	Independent	Republican
Democrat	284	123	166
Independent	0	0	0
Republican	96	116	159

We can compute the proportion correctly classified as:

```
(284+0+159)/nrow(rnes96)
```

```
[1] 0.46928
```

We see that only 47% of the current data are correctly classified and we could expect the performance on new individuals to be slightly worse than this. No cases are classified as independents because the probability of the other two outcomes always dominates this outcome. We see that the majority of actual Republicans are classified as Democrats. So the performance is not impressive but given the large overlap in the predictor space between the three levels (or classes), it is not surprising.

The multinomial logit model is not usually the best choice for classification performance. Better performance can be obtained by methods specialized for this purpose such as random forests or support vector machines. Even so, the multinomial logit model often provides better insight into how and which predictors affect the classification. This understanding of how a classification method works can be invaluable in predicting how it will behave in new circumstances or how predictors might be added or modified. The more sophisticated methods are often opaque as to meaning so while they may perform well, they may add little to our understanding.

We can examine the coefficients to gain an understanding of the relationship between the predictor and the response:

```
summary(mmodi)
```

Coefficients:

```
(Intercept)  nincome
```

```
Independent    -1.17493 0.016087
Republican     -0.95036 0.017665
```

Std. Errors:

```
              (Intercept)    nincome
Independent    0.15361 0.0028497
Republican     0.14169 0.0026525
```

Residual Deviance: 1985.4

AIC: 1993.4

The intercept terms model the probabilities of the party identification for an income of zero. We can see the relationship from this calculation:

```
cc <- c(0, -1.17493, -0.95036)
exp(cc) / sum(exp(cc))
[1] 0.58982 0.18216 0.22802
predict(mmodi, data.frame(income=0), type="probs")
      Democrat Independent Republican
      0.58982      0.18216      0.22802
```

The slope terms represent the log-odds of moving from the baseline category of Democrat to Independent and Republican, respectively, for a unit change of \$1000 in income. We can see more explicitly what this means by predicting probabilities for incomes \$1000 apart and then computing the log-odds:

```
(pp <- predict(mmodi, data.frame(income=c(0, 1)), type="probs"))
      Democrat Independent Republican
1  0.58982      0.18216      0.22802
2  0.58571      0.18382      0.23047
log(pp[1, 1] * pp[2, 2] / (pp[1, 2] * pp[2, 1]))
[1] 0.016087
log(pp[1, 1] * pp[2, 3] / (pp[1, 3] * pp[2, 1]))
[1] 0.017665
```

Log-odds can be difficult to interpret particularly with many predictors and interactions. Sometimes, computing predicted probabilities for a selected range of predictors can provide better intuition.

7.2 Linear Discriminant Analysis

Linear discriminant analysis (LDA) is a method based on linear combinations of the predictors that classifies cases into one of a fixed number of possibilities. This prediction is called *classification* and LDA is usually considered as a classification method. A multinomial logit model explains the relationship between a response that can take one of a fixed number of levels and linear combinations of the predictors. Although LDA focuses more on prediction and the multinomial logit more on explanation, we can learn from comparing the two. We give only a brief introduction to LDA here and the reader is encouraged to consult other sources regarding this method. We focus on the explanatory insights offered by LDA in this section.

Consider the vector of predictors $\mathbf{x}_i$ for $i = 1, \dots, n$. We can compute a measure of total variation in the data using the total sum of squares S :

$$S = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

Suppose we have G groups and we add a group subscript g to denote which group a case belongs to. We define the within-group covariance W as:

$$W = \sum_{g=1}^G \sum_{i=1}^{n_g} (\mathbf{x}_{gi} - \bar{\mathbf{x}}_g)(\mathbf{x}_{gi} - \bar{\mathbf{x}}_g)^T$$

We define the between-groups covariance B as:

$$B = \sum_{g=1}^G (\bar{\mathbf{x}}_g - \bar{\mathbf{x}})(\bar{\mathbf{x}}_g - \bar{\mathbf{x}})^T$$

where n_g is the number of cases in group g . We have $S = W + B$. We want to form linear combinations $\mathbf{a}$ of the predictors to maximize the separation between the groups. To achieve this, we choose $\mathbf{a}$ to maximize:

$$\frac{\mathbf{a}^T B \mathbf{a}}{\mathbf{a}^T W \mathbf{a}}$$

The solution can be found using eigendecomposition of $W^{-1}B$. The first eigenvector represents the combination which maximizes the ratio with subsequent eigenvectors representing the next best solutions subject to the requirement that they are orthogonal to previous eigenvectors.

The calculation can be performed using the MASS library.

```
library(MASS)
mllda <- lda(party ~ age + income, rnes96)
mllda
```

```
Prior probabilities of groups:
      Democrat Independent  Republican
      0.40254      0.25318      0.34428
```

```
Group means:
      age income
Democrat  47.066 37.632
Independent 46.506 51.655
Republican  47.412 53.298
```

```
Coefficients of linear discriminants:
      LD1      LD2
age    0.0053257 0.060821
income 0.0332155 -0.000114
```

```
Proportion of trace:
      LD1      LD2
0.9926 0.0074
```

The prior probabilities are chosen by default as the observed proportion in the data. One can specify the prior if, for example, we have a biased sample in which the class sizes do not correspond to the population. The means in the groups, $\bar{x}_g$, reveal that there is not much difference in the ages of the three groups but there are noticeable differences in income. The proportion of the trace is computed using the eigenvalues of the decomposition. In this case, we see that the first component is strongly dominant and so the classification will depend mostly on this. The coefficients give us

the **a**. We see that the first combination is dominated by the income while the second by the age. However, the second combination counts for little so the classification is based mainly on the income.

We can use the LDA to classify the current data:

```
preds <- predict(mlda)
head(preds$posterior)
```

	Democrat	Independent	Republican
1	0.58666	0.18945	0.22389
2	0.59537	0.19135	0.21328
3	0.59322	0.19089	0.21590
4	0.59105	0.19042	0.21854
5	0.56846	0.18533	0.24620
6	0.59483	0.19124	0.21393

The posterior probabilities computed from the model can be used to classify the cases. The first six cases are all classified as Democrats as this is highest probability in each case.

We can get the most likely outcome from each case and compare it against the observed class:

```
xtabs( ~ predict(mlda)$class + rnes96$party)
```

	rnes96\$party		
predict(mlda)\$class	Democrat	Independent	Republican
Democrat	298	141	184
Independent	0	0	0
Republican	82	98	141

The result is quite similar to the multinomial logit in that no cases are classified as Independent and many true Republicans are classified as Democrats.

7.3 Hierarchical or Nested Responses

Consider the following data collected by Lowe et al. (1971) by way of McCullagh and Nelder (1989) concerning live births with deformations of the central nervous system in south Wales:

```
data(cns, package="faraway")
cns
```

	Area	NoCNS	An	Sp	Other	Water	Work
1	Cardiff	4091	5	9	5	110	NonManual
2	Newport	1515	1	7	0	100	NonManual
3	Swansea	2394	9	5	0	95	NonManual
4	GlamorganE	3163	9	14	3	42	NonManual
5	GlamorganW	1979	5	10	1	39	NonManual
6	GlamorganC	4838	11	12	2	161	NonManual
7	MonmouthV	2362	6	8	4	83	NonManual
8	MonmouthOther	1604	3	6	0	122	NonManual
9	Cardiff	9424	31	33	14	110	Manual
10	Newport	4610	3	15	6	100	Manual
11	Swansea	5526	19	30	4	95	Manual
12	GlamorganE	13217	55	71	19	42	Manual
13	GlamorganW	8195	30	44	10	39	Manual
14	GlamorganC	7803	25	28	12	161	Manual
15	MonmouthV	9962	36	37	13	83	Manual
16	MonmouthOther	3172	8	13	3	122	Manual

NoCNS indicates no central nervous system (CNS) malformation. An denotes anencephalus while Sp denotes spina bifida and Other represents other malformations. Water is water hardness and the subjects are categorized by the type of work performed by the parents. We might consider a multinomial response with four categories. However, we can see that most births suffer no malformation and so this category dominates the other three. It is better to consider this as a hierarchical response as depicted in Figure 7.3. Now consider the multinomial likelihood for the i^{th}

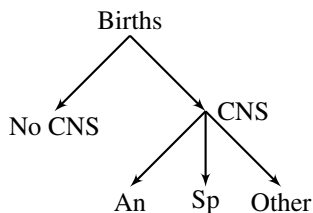


Figure 7.3 Hierarchical response for birth types.

observation which is proportional to:

$$p_{i1}^{y_{i1}} p_{i2}^{y_{i2}} p_{i3}^{y_{i3}} p_{i4}^{y_{i4}}$$

Define $p_{ic} = p_{i2} + p_{i3} + p_{i4}$ which is probability of a birth with some kind of CNS malformation. We can then write the likelihood as:

$$p_{i1}^{y_{i1}} p_{ic}^{y_{i2}+y_{i3}+y_{i4}} \times \left(\frac{p_{i2}}{p_{ic}} \right)^{y_{i2}} \left(\frac{p_{i3}}{p_{ic}} \right)^{y_{i3}} \left(\frac{p_{i4}}{p_{ic}} \right)^{y_{i4}}$$

The first part of the product is now a binomial likelihood for a CNS vs. NoCNS response. The second part of the product is now a multinomial likelihood for the three CNS categories conditional of the presence of CNS. For example, p_{i2}/p_{ic} is the conditional probability of an anencephalus birth given that a malformation has occurred for the i^{th} observation. We can now separately develop a binomial model for whether malformation occurs and a multinomial model for the type of malformation.

We start with the binomial model. First we accumulate the number of CNS births and plot the data with the response on the logit scale as shown in the first panel of Figure 7.4:

```
cns$CNS <- cns$An+cns$Sp+cns$Other
plot(log(CNS/NoCNS) ~ Water, cns, pch=as.character(Work))
```

We observe that the proportion of CNS births falls with increasing water hardness and is higher for manual workers. We observe one observation (manual, Newport) that may be an outlier. Notice that the Area is confounded with the Water hardness, so we cannot put both these predictors in our model. But Water does represent a subspace of Area so it is legitimate to compare:

```
binmodw <- glm(cbind(CNS,NoCNS) ~ Water + Work, cns, family=binomial)
binmoda <- glm(cbind(CNS,NoCNS) ~ Area + Work, cns, family=binomial)
anova(binmodw,binmoda,test="Chi")
```

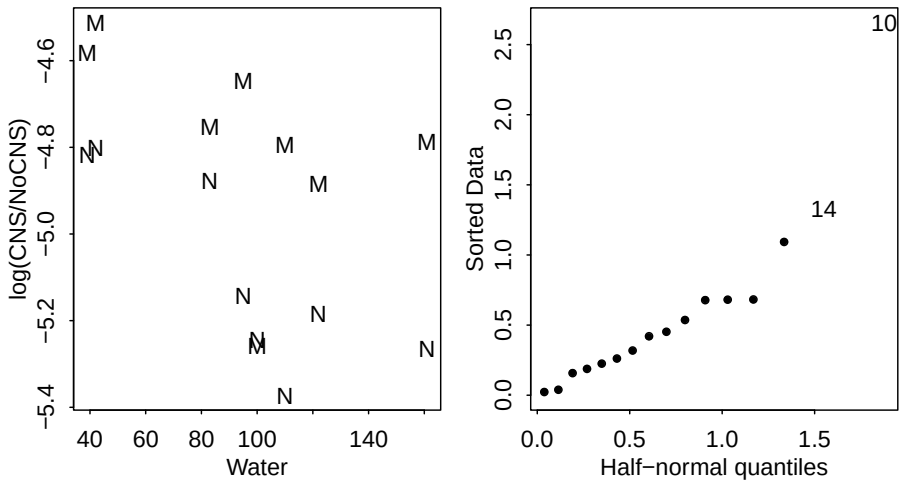


Figure 7.4 The first plot shows the empirical logits for the proportion of CNS births related to water hardness and profession (M=Manual, N=Nonmanual). The second is a half-normal plot of the residuals of the chosen model.

Analysis of Deviance Table

```
Model 1: cbind(CNS, NoCNS) ~ Water + Work
Model 2: cbind(CNS, NoCNS) ~ Area + Work
  Resid. Df Resid. Dev Df Deviance P(>|Chi|)
1      13      12.36
2       7       3.08  6     9.29    0.16
```

One can view this test as a check for linear trend in the effect of water hardness. We find that the simpler model using `Water` is acceptable. A check for an interaction effect revealed nothing significant although a look at the residuals is worthwhile:

```
library(faraway)
halfnorm(residuals(binmodw))
```

In the second plot of Figure 7.4, we see an outlier corresponding to Newport manual workers. This case deserves closer examination. Finally, a look at the chosen model:

```
summary(binmodw)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.432580	0.089789	-49.37	< 2e-16
Water	-0.003264	0.000968	-3.37	0.00075
WorkNonManual	-0.339058	0.097094	-3.49	0.00048

`n = 16 p = 3`

Deviance = 12.363 Null Deviance = 41.047 (Difference = 28.685)

The residual deviance is close to the degrees of freedom indicating a reasonable fit to the data. We see that since:

```
exp(-0.339058)
```

```
[1] 0.71244
```

births to nonmanual workers have a 29% lower chance of CNS malformation. Water hardness ranges from about 40 to 160. So a difference of 120 would decrease the odds of CNS malformation by about 32%.

Now consider a multinomial model for the three malformation types conditional on a malformation having occurred. As this data is grouped, in contrast to the `nes96` example, it is most convenient to present the response as a matrix:

```
cmmod <- multinom(cbind(An,Sp,Other) ~ Water + Work, cns)
```

We find that neither predictor has much effect:

```
nmod <- step(cmmod)
```

```
      Df    AIC
- Water  4 1381.1
- Work   4 1381.2
<none>   6 1383.5
```

```
      Df    AIC
- Work   2 1378.5
<none>   4 1381.1
```

which leaves us with a null final model:

```
nmod
```

Coefficients:

```
      (Intercept)
Sp              0.28963
Other          -0.98083
```

Residual Deviance: 1374.5

The fitted proportions are:

```
cc <- c(0, 0.28963, -0.98083)
```

```
names(cc) <- c("An", "Sp", "Other")
```

```
exp(cc)/sum(exp(cc))
```

```
      An      Sp      Other
0.36888 0.49279 0.13833
```

So we find that water hardness and parents' professions are related to the probability of a malformed birth, but that they have no effect on the type of malformation.

Observe that if we fit a multinomial logit model to all four categories:

```
multinom(cbind(NoCNS,An,Sp,Other) ~ Water + Work, cns)
```

Coefficients:

```
      (Intercept)      Water WorkNonManual
An      -5.4551 -0.00290884      -0.36388
Sp      -5.0710 -0.00432305      -0.24359
Other   -6.5947 -0.00051358      -0.64219
```

Residual Deviance: 9391

AIC: 9409

We find that both `Water` and `Work` are significant, but that the fact that they do not distinguish the type of malformation is not easily discovered from this model.

7.4 Ordinal Multinomial Responses

Suppose we have J ordered categories and that for individual i , with ordinal response Y_i , $p_{ij} = P(Y_i = j)$ for $j = 1, \dots, J$. With an ordered response, it is often easier to work with the cumulative probabilities, $\gamma_{ij} = P(Y_i \leq j)$. The cumulative probabilities are increasing and invariant to combining adjacent categories. Furthermore, $\gamma_{iJ} = 1$, so we need only model $J - 1$ probabilities.

As usual, we must link the γ s to the covariates x . We will consider three possibilities which all take the form:

$$g(\gamma_{ij}) = \theta_j - x_i^T \beta$$

Possible link functions, g , are the logit, the probit and the complementary log-log. We have explicitly specified the intercepts, θ_j , so that the vector x_i does not include an intercept. Furthermore, β does not depend on j so that we assume that the predictors have a uniform effect on the response categories in a sense that we will shortly make clear.

Suppose that Z_i is some unobserved continuous variable that might be thought of as the real underlying latent response. We only observe a discretized version of Z_i in the form of Y_i where $Y_i = j$ is observed if $\theta_{j-1} < Z_i \leq \theta_j$. Further suppose that $Z_i - \beta^T x_i$ has distribution F , then:

$$P(Y_i \leq j) = P(Z_i \leq \theta_j) = P(Z_i - \beta^T x_i \leq \theta_j - \beta^T x_i) = F(\theta_j - \beta^T x_i)$$

Now if, for example, F follows the logistic distribution, where $F(x) = e^x / (1 + e^x)$, then:

$$\gamma_{ij} = \frac{\exp(\theta_j - \beta^T x_i)}{1 + \exp(\theta_j - \beta^T x_i)}$$

and so we would have a logit model for the cumulative probabilities γ_{ij} . Choosing the normal distribution for the latent variable leads to a probit model, while the choice of an extreme value distribution leads to the complementary log-log. This *latent* variable explanation for the model is displayed in Figure 7.5. We include the R code for reference.

```
x <- seq(-5, 5, by=0.1)
xa <- c(21, 41, 71)
plot(x, dlogis(x), type="l", axes=FALSE, ylab="", xlab="")
segments(x[xa], rep(0, 3), x[xa], dlogis(x[xa]))
axis(1, at=x[xa], c(expression(theta[1]), expression(theta[2]), expression(theta[3])))
```

Notice that if $\beta > 0$, as x_i increases, $P(Y_i = J)$ will also increase. This explains the use of the minus sign in the definition of the model because it allows for the more intuitive interpretation of the sign of β .

Proportional Odds Model: Let $\gamma_j(x_i) = P(Y_i \leq j | x_i)$, then the *proportional odds* model, which uses the logit link, is:

$$\log \frac{\gamma_j(x_i)}{1 - \gamma_j(x_i)} = \theta_j - \beta^T x_i, \quad j = 1, \dots, J - 1$$

It is called this because the relative odds for $Y \leq j$ comparing x_1 and x_2 are:

$$\left(\frac{\gamma_j(x_1)}{1 - \gamma_j(x_1)} \right) / \left(\frac{\gamma_j(x_2)}{1 - \gamma_j(x_2)} \right) = \exp(-\beta^T (x_1 - x_2))$$

This does not depend on j . Of course, the assumption of proportional odds does need to be checked for a given dataset.

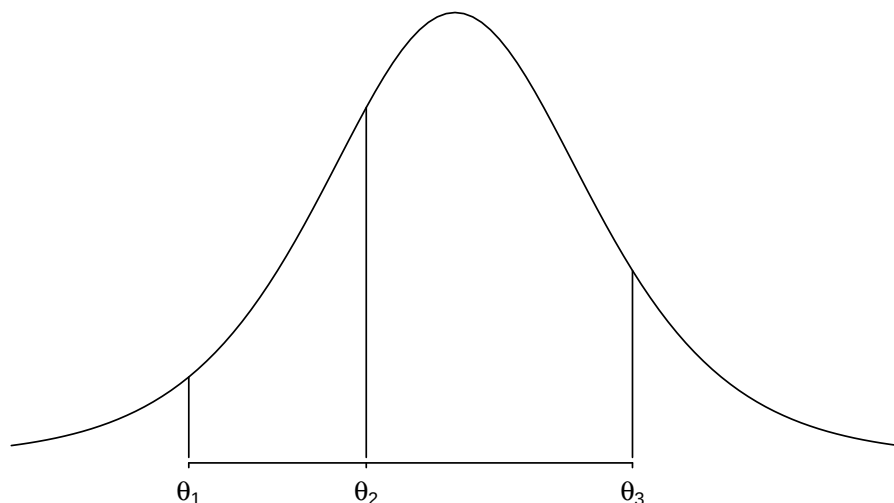


Figure 7.5 *Latent variable view of an ordered multinomial response. Here, four discrete responses can occur, depending on the position of Z relative to the cutpoints θ_j . As x changes, the cutpoints will move together to change the relative probabilities of the four responses.*

Returning to the `nes96` dataset, suppose we assume that Independents fall somewhere between Democrats and Republicans. We would then have an ordered multinomial response. We can then fit this using the `polr` function from the `MASS` library described in Venables and Ripley (2002):

```
library(MASS)
pomod <- polr(party ~ age + education + income, rnes96)
```

The deviance and number of parameters for this model are:

```
c(deviance(pomod), pomod$edf)
```

```
[1] 1984.2 10.0
```

which can be compared to the corresponding multinomial logit model:

```
c(deviance(mmod), mmod$edf)
```

```
[1] 1968.3 18.0
```

The proportional odds model uses fewer parameters, but does not fit quite as well. Typically, the output from the proportional odds model is easier to interpret. We may use an AIC-based variable selection method:

```
pomodi <- step(pomod)
```

```
Start: AIC=2004.2
```

```
party ~ age + education + income
```

```

      Df AIC
- education 6 2003
<none>      2004
- age       1 2004
- income    1 2039
```

```
Step: AIC=2002.8
```

```
party ~ age + income
```

```

      Df  AIC
- age    1 2001
<none>   2003
- income 1 2047

```

```

Step:  AIC=2001.4
party ~ income

```

```

      Df  AIC
<none>   2001
- income 1 2045

```

Thus we finish with a model including just income as we did with the earlier multinomial model. We could also use a likelihood ratio test to compare the models:

```

deviance(pomod) - deviance(pomod)
[1] 11.151
pchisq(11.151, pomod$edf - pomodi$edf, lower=F)
[1] 0.13217

```

We see that the simplification to just income is justifiable. We can check the proportional odds assumption by computing the observed odds proportions with respect to, in this case, income levels. We have computed the log-odds difference between γ_1 and γ_2 :

```

pim <- with(rnes96, prop.table(table(income, party), 1))
logit(pim[, 1]) - logit(pim[, 1] + pim[, 2])

```

	1.5	4	6	8	9.5	10.5	11.5
-0.90079	-2.06142	-0.75769	-1.00330	-2.30259	-0.30830	-0.79851	
12.5	13.5	14.5	16	18.5	21	23.5	
-1.89712	-1.25276	-1.17865	-0.41285	-0.35424	-1.51413	-1.65345	
27.5	32.5	37.5	42.5	47.5	55	67.5	
-0.74678	-0.52252	-0.92326	-1.02962	-0.82198	-1.42760	-1.18261	
82.5	97.5	115					
-0.98676	-1.48292	-1.70660					

It is questionable whether these can be considered sufficiently constant, but at least there is no trend. Now consider the interpretation of the fitted coefficients:

```

summary(pomod)
Coefficients:
      Value Std. Error t value
income 0.013120  0.0019708  6.6572

```

```

Intercepts:
      Value Std. Error t value
Democrat|Independent  0.209  0.112    1.863
Independent|Republican 1.292  0.120   10.753

```

```

Residual Deviance: 1995.36
AIC: 2001.36

```

We can say that the odds of moving from Democrat to Independent/Republican category (or from Democrat/Independent to Republican) increase by a factor of $\exp(0.013120) = 1.0132$ as income increases by one unit (\$1000). Notice that the log-odds are similar to those obtained in the multinomial logit model. The intercepts correspond to the θ_j . So for an income of \$0, the predicted probability of being a Democrat is:

```

ilogit(0.209)

```

```
[1] 0.55206
while that of being an Independent is:
ilogit(1.292)-ilogit(0.209)
[1] 0.23242
with the remainder being Republicans. We can compute predicted values:
inclevels <- seq(0,100,by=20)
predict(pomodi,data.frame(income=inclevels,row.names=inclevels), type=
  ↪ "probs")
```

	Democrat	Independent	Republican
0	0.55209	0.23232	0.21559
20	0.48668	0.25007	0.26325
40	0.42173	0.26109	0.31718
60	0.35937	0.26411	0.37652
80	0.30143	0.25877	0.43980
100	0.24920	0.24569	0.50511

Notice how the probability of being a Democrat uniformly decreases with income while that for being a Republican uniformly increases as income increases, but that the middle category of Independent increases then decreases. This type of behavior can be expected from the latent variable representation of the model.

We can illustrate the latent variable interpretation of proportional odds by computing the cutpoints for incomes of \$0, \$50,000 and \$100,000:

```
x <- seq(-4,4,by=0.05)
plot(x,dlogis(x),type="l")
abline(v=c(0.209,1.292))
abline(v=c(0.209,1.292)-50*0.013120, lty=2)
abline(v=c(0.209,1.292)-100*0.013120, lty=5)
```

The plot is shown in Figure 7.6.

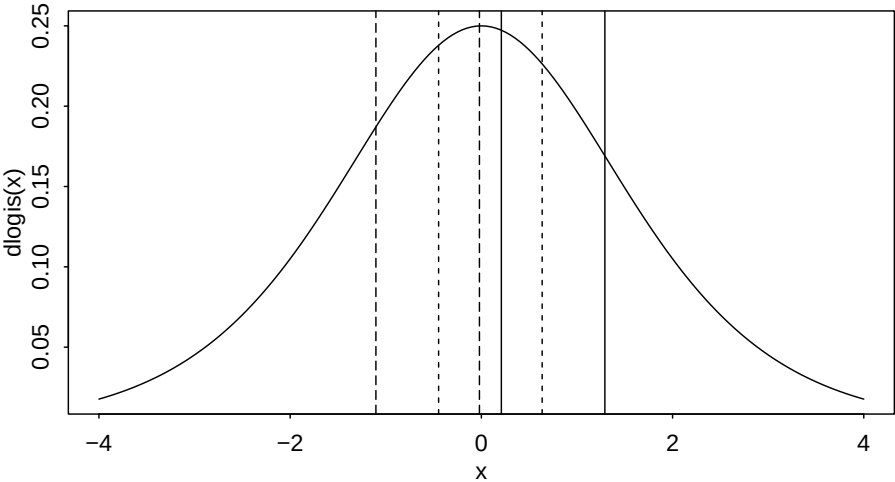


Figure 7.6 Solid lines represent an income of \$0, dotted lines are for \$50,000 and dashed lines are for \$100,000. The probability of being a Democrat is given by the area lying to the left of the leftmost of each pair of lines, while the probability of being a Republican is given by the area to the right of the rightmost of the pair. Independents are represented by the area in-between.

Generalization: The proportional odds models can be generalized by allowing β to vary by

$$\log \frac{\gamma_j(x)}{1 - \gamma_j(x)} = \theta_j - \beta_j^T x \quad j = 1, \dots, k-1$$

but this loses the proportionality property. Such models cannot be fitted using `polr()` but are provided for in the VGAM package of Yee (2010). Here's how we fit such a model to the current data:

```
library(VGAM)
nmod <- vglm(party ~ income, family=cumulative(parallel=FALSE), rnes96)
summary(nmod)
```

```
Coefficients:
              Estimate Std. Error z value
(Intercept):1    0.3289    0.12156   2.71
(Intercept):2    1.1483    0.12872   8.92
income:1         -0.0162    0.00236  -6.86
income:2         -0.0105    0.00220  -4.76
```

Residual deviance: 1987.5 on 1884 degrees of freedom

The parameterization is different for this package in that $\theta_j + \beta_j^T x$ is used for the linear predictor replacing the $-$ with a $+$ in the `polr` form. The argument `parallel = FALSE` indicates we have chosen the nonproportional (or parallel) option.

We compare this to the proportional odds model and use the standard likelihood theory to compare the models:

```
pmod <- vglm(party ~ income, family=cumulative(parallel=TRUE), rnes96)
deviance(pmod) - deviance(nmod)
[1] 7.8241
1-pchisq(7.82,1)
[1] 0.0051671
```

There is a difference of only one parameter between the two models. We see that the simplification to the proportional odds model is not justified here because the p -value is small. However, we should bear in mind that we have a fairly large sample size of almost 1000 cases so even relatively small differences are prone to be significant. So our previous use of the proportional odds model was not completely unreasonable.

Predictions can be made in a straightforward manner. Interpreting the model is somewhat more difficult due to the varying slopes. Nonparallel lines will cross somewhere. For this model that point is:

```
(1.148-0.329)/(0.0105-0.0162)
[1] -143.68
```

So at an income below $-\$144,000$, the predicted probability of being Democrat would exceed the probability of being Democrat or Independent. Clearly this is impossible and the model predictions are ridiculous. This is not a practical problem in this example as $-\$144,000$ is far outside the range of incomes we would consider. But in other situations, particularly where more predictors are used, this could become a serious issue. This is one good reason to prefer the proportional model if it is viable.

Ordered Probit Model: If the latent variable Z_i has a standard normal distribution, then:

$$\Phi^{-1}(\gamma_j(x_i)) = \theta_j - \beta_j^T x_i \quad j = 1, \dots, J-1$$

Applying this model to the `nes96` data, we find:

```
opmod <- polr(party ~ income, method="probit", rnes96)
summary(opmod)
```

Coefficients:

	Value	Std. Error	t value
nincome	0.008182	0.0012078	6.7745

Intercepts:

	Value	Std. Error	t value
Democrat Independent	0.128	0.069	1.851
Independent Republican	0.798	0.072	11.040

Residual Deviance: 1994.89

AIC: 2000.89

The deviance is similar to the logit version of this model, but the coefficients appear to be different. However, if we compute the same predictions:

```
dems <- pnorm(0.128-inclevels*0.008182)
demind <- pnorm(0.798-inclevels*0.008182)
data.frame(Democrat=dems, Independent=demind-dems, Republican=1-demind,
  <- row.names=inclevels)
```

	Democrat	Independent	Republican
0	0.55093	0.23664	0.21244
20	0.48578	0.25129	0.26292
40	0.42102	0.26006	0.31892
60	0.35833	0.26228	0.37939
80	0.29925	0.25778	0.44297
100	0.24503	0.24691	0.50806

We see that the predicted values are very similar to those seen for the logit. If the coefficients are appropriately rescaled, they are also very similar.

Proportional Hazards Model: A concept of *hazard* was developed in insurance applications. When issuing a life insurance policy, the insurer is interested in the probability that the person will die during the term of the policy given that they are alive now. This is not the same as the unconditional probability of death during the same time period. In other words, for example, we want to know the chance that a 55-year-old man will die in the next year, given that he is alive and aged 55. The unconditional probability that a man will die aged 55 is not particularly useful for the purposes of insurance.

Suppose we use the complementary log-log in place of the logit above, that is:

$$\log(-\log(1 - \gamma_j(x_i))) = \theta_j + \beta^T x_i$$

Then the *hazard* of category j is the probability of falling in category j given that your category is greater than j :

$$\text{Hazard}(j) = P(Y_i = j | Y_i \geq j) = \frac{P(Y_i = j)}{P(Y_i \geq j)} = \frac{p_j}{1 - \gamma_{i,j-1}} = \frac{\gamma_{ij} - \gamma_{i,j-1}}{1 - \gamma_{i,j-1}}$$

The corresponding latent variable distribution is the extreme value:

$$F(x) = 1 - \exp(-\exp(x))$$

The extreme value distribution is not symmetric like the logistic and normal and so there seems little justification for applying it to the `nes96` data, but the command is:

```
polr(party ~ income, method="cloglog")
```

Assigning Scores: When the ordinal response has a larger number of categories, it may be sensible to assign scores to each level and then model these scores as the response in a standard linear model. Suppose we reconsider the `nes96` data but use the income as the response variable. In the original form of the data this is an ordinal variable:

```
levels(nes96$income)
```

```
[1] "$3Kminus" "$3K-$5K" "$5K-$7K" "$7K-$9K" "$9K-$10K"
[6] "$10K-$11K" "$11K-$12K" "$12K-$13K" "$13K-$14K" "$14K-$15K"
[11] "$15K-$17K" "$17K-$20K" "$20K-$22K" "$22K-$25K" "$25K-$30K"
[16] "$30K-$35K" "$35K-$40K" "$40K-$45K" "$45K-$50K" "$50K-$60K"
[21] "$60K-$75K" "$75K-$90K" "$90K-$105K" "$105Kplus"
```

This is a special case of ordinal variable called an *interval* variable because a continuous variable has been discretized into intervals. For such variables, it is natural to assign scores equal to the midpoints of the intervals:

```
inca <- c(1.5, 4, 6, 8, 9.5, 10.5, 11.5, 12.5, 13.5, 14.5, 16, 18.5, 21, 23.5,
  ↪ 27.5, 32.5, 37.5, 42.5, 47.5, 55, 67.5, 82.5, 97.5, 115)
nes96$sincome <- inca[unclass(nes96$income)]
```

The `unclass` function converts a categorical variable into an integer variable with the levels numbered in the order in which they appear. We have to choose a score for the first and last levels. It is natural to suppose that zero is the lower bound giving a choice of 1.5 for the first interval. We have chosen 115 for the last interval but other choices may be reasonable. For noninterval valued ordinal variables the choice of scores is much more subjective and should reflect an understanding of the relative difference between adjacent levels.

Now we can fit a linear model using the same variables considered previously (but using the original form of the party identification variable):

```
lmmod <- lm(sincome ~ age + educ + PID, nes96)
summary(lmmod)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	45.8380	3.3533	13.67	< 2e-16
age	-0.0496	0.0582	-0.85	0.394
educ.L	39.2854	5.0850	7.73	2.9e-14
educ.Q	2.8554	4.7508	0.60	0.548
educ.C	3.4486	4.0083	0.86	0.390
educ^4	2.5299	3.2825	0.77	0.441
educ^5	-4.4705	2.7992	-1.60	0.111
educ^6	-0.5773	2.3419	-0.25	0.805
PID.L	12.7103	2.1634	5.88	5.9e-09
PID.Q	-6.3140	2.8613	-2.21	0.028
PID.C	1.0306	2.3740	0.43	0.664
PID^4	6.2413	2.9296	2.13	0.033
PID^5	-2.5174	2.5585	-0.98	0.325
PID^6	-0.7923	3.6532	-0.22	0.828

```
n = 944, p = 14, Residual SE = 27.899, R-Squared = 0.2
```

Both the education and party identification variable are ordinal and so are represented using polynomial contrasts. These can be hard to interpret when we have several levels. One option is to recode these predictors as nominal factors which is more straightforward to understand but loses some information. In this case, we notice that the linear term for both education and party identification is the only strongly

significant term. This provides the hint that we might simplify this model by also assigning scores to these two predictors. We also notice that age appears not to be needed. We can then compare this simplified model to the original model:

```
lmod2 <- lm(sincome ~ unclass(educ) + unclass(PID), nes96)
anova(lmod2, lmod)
```

Analysis of Variance Table

Model 1: sincome ~ unclass(educ) + unclass(PID)

Model 2: sincome ~ age + educ + PID

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	941	737108				
2	930	723881	11	13227	1.54	0.11

We see that the simplification is justifiable. We need to know the ordering of the scores before interpreting the model:

```
levels(nes96$educ)
```

```
[1] "MS" "HSDrop" "HS" "Coll" "CCdeg" "BAdeg" "MAdeg"
```

```
levels(nes96$PID)
```

```
[1] "strDem" "weakDem" "indDem" "indind" "indRep" "weakRep"
```

```
[7] "strRep"
```

Now we consider the model summary:

```
summary(lmod2)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.510	3.038	1.16	0.25
unclass(educ)	7.359	0.573	12.84	< 2e-16
unclass(PID)	2.463	0.403	6.11	1.4e-09

n = 944, p = 3, Residual SE = 27.988, R-Squared = 0.19

We see that each one-step increase in the level of education is associated with a \$7359 increase in income. We also see that each step along our political scale from strongly Democrat to strongly Republican is associated with \$2463 increase income. This association should only be viewed predictively and not causally. It would certainly be surprising if changing your political opinions caused you to earn more or less. Furthermore, more intelligent and/or privileged people would tend to earn more regardless of their level of formal education.

Further Reading: For more on the analysis of ordered categorical data see the books by Agresti (1984), Clogg and Shihadeh (1994), Powers and Xie (2000) and Simonoff (2003).

Exercises

1. The `hsb` data was collected as a subset of the High School and Beyond study conducted by the National Education Longitudinal Studies program of the National Center for Education Statistics. The variables are gender; race; socioeconomic status (SES); school type; chosen high school program type; scores on reading, writing, math, science, and social studies. We want to determine which factors are related to the choice of the type of program — academic, vocational or general — that the students pursue in high school. The response is multinomial with three levels.
 - (a) Make a table showing the proportion of males and females choosing the three

different programs. Comment on the difference. Repeat this comparison but for SES rather than gender.

- (b) Construct a plot like the right panel of Figure 7.1 that shows the relationship between program choice and reading score. Comment on the plot. Repeat for math in place of reading.
 - (c) Compute the correlation matrix for the five subject scores.
 - (d) Fit a multinomial response model for the program choice and examine the fitted coefficients. Of the five subjects, one gives unexpected coefficients. Identify this subject and suggest an explanation for this behavior.
 - (e) Construct a derived variable that is the sum of the five subject scores. Fit a multinomial model as before except with this one sum variable in place of the five subjects separately. Compare the two models to decide which should be preferred.
 - (f) Use a stepwise method to reduce the model. Which variables are in your selected model?
 - (g) Construct a plot of predicted probabilities from your selected model where the math score varies over the observed range. Other predictors should be set at the most common level or mean value as appropriate. Your plot should be similar to Figure 7.2. Comment on the relationship.
 - (h) Compute a table of predicted probabilities cross-classified by SES and school type. Fix the other predictors at their mean values. Comment on how SES and school type are related to the response.
 - (i) Compute the predicted outcome for the student with ID 99. What did this student actually choose?
 - (j) Construct a table of the most likely predicted outcomes and observed outcomes. In what proportion of cases is the predicted and observed outcome the same?
2. Data were collected from 39 students in a University of Chicago MBA class and may be found in the dataset `happy`.
- (a) Make plots of the relationship between the response, happiness and each of the four predictors. Transform sex and love to appropriate factors.
 - (b) Fit a proportional odds model with the main effects of each of the four predictors. Extract the estimated values of θ using the zeta component of the fit. Make an index plot of these and comment on any deviation from linearity and what this may signify.
 - (c) Use stepwise AIC to select a reduced model. Which variables are eliminated. Make a test to compare this reduced model to the original model.
 - (d) Compare a model where love is treated as an ordered factor with one where it is treated as a numerical predictor. Which model is preferred?
 - (e) Examine the regression coefficients of your selected model and provide an interpretation of their meaning.

- (f) Varying money across its range and holding love and work fixed at their median values, predict the most likely happiness level. Repeat the process varying only love and then work. What do we learn about the effect of the three predictors on the response?
 - (g) Predict the happiness distribution for subject whose parents earn \$30,000 a year, who is lonely, not sexually active and has no job.
3. A student newspaper conducted a survey of student opinions about the Vietnam War in May 1967. Responses were classified by sex, year in the program and one of four opinions. The survey was voluntary. The data may be found in the dataset `uncviet`.
- (a) Compute the proportion favoring each policy within each year by sex combination. Plot these proportions as year varies with a different line type for each policy. Plot men and women on separate panels. Your plot should look similar to Figure 7.1.
 - (b) Fit a proportional odds model with policy as the response with sex and year as predictors (include their interaction). Use the `weight` argument to set the number of respondents for each case. Why is it sensible to include an interaction term?
 - (c) Now fit a model with main effects only, excluding the interaction. Compare this model to the previous one. Which is preferred?
 - (d) Compute the predicted proportion for each case and plot in the same format as (a). Use the `type="probs"` argument to predict. You will need to select only one of the four probabilities for each case. Comment on the plot and compare it to (a).
 - (e) Compute the raw residuals as the difference between the predicted and observed proportions. Use the same format to plot these residuals as the predicted proportions. Comment on the plot and suggest how the definition of the residual might be improved.
 - (f) Examine the regression summary output to find the coefficients that are relevant to the opinions of women. What does the significance (or lack thereof) say about how the opinions of women vary across the year groups?
4. The `pneumo` data gives the number of coal miners classified by radiological examination into one of three categories of pneumoconiosis and by the number of years spent working at the coal face divided into eight categories.
- (a) Make a plot showing how the proportion on miners in the three categories at each year point varies over time. Comment on the relationship.
 - (b) Treating the pneumoconiosis status as response variable as nominal, build a model for predicting the frequency of the three outcomes in terms of length of service. What does the model say about the similarity of the proportions falling into the mild and severe categories?
 - (c) Would it be better to use `log(year)` as the predictor?
 - (d) Produce a plot of the predicted probabilities in the same format as (a).

- (e) Fit a proportional odds model to the data. Take care to order response correctly. You will need to specify the number falling in each case using the `weight` argument. What is the estimated value of θ_1 and how should it be interpreted?
 - (f) Repeat the analysis with the pneumoconiosis status being treated as ordinal.
 - (g) Extract the predicted probabilities from the model and plot in the same format as (a). Compare to the predictions from the nominal model.
 - (h) Fit a hierarchical model to the status response. First fit a binomial response—normal and not normal. Interpret the effect of year on the odds of getting the lung disease.
 - (i) Now fit a binomial model for mild vs. severe lung disease. Is the year effect significant. What is the probability of a mild disease within the diseased subgroup?
 - (j) Compute the predicted probabilities of the three categories by combining the two binomial model predictions. Plot in the same format as (a) and comment.
5. The `debt` data arise from a large postal survey on the psychology of debt. The frequency of credit card use is a three-level factor ranging from never, through occasionally to regularly.
- (a) Declare the response as an ordered factor and make a plot showing the relationship to `prodebt`. Comment on the plot. Use a table or plot to display the relationship between the response and the income group.
 - (b) Fit a proportional odds model for credit card use with all the other variables as predictors. What are the two most significant predictors (largest t -values) and what is their qualitative effect on the response? What is the least significant predictor?
 - (c) Fit a proportional odds model using only the least significant predictor from the previous model. What is the significance of this predictor in this small model? Are the conclusions regarding this predictor contradictory for the two models?
 - (d) Use stepwise AIC to select a smaller model than the full set of predictors. You will need to handle the missing values carefully. Report on the qualitative effect of the predictors in your chosen model. Can we conclude that the predictors that were dropped from the model have no relation to the response?
 - (e) Compute the median values of the predictors in your selected model. At these median values, contrast the predicted outcome probabilities for both smokers and nonsmokers.
 - (f) Fit a proportional hazards model to the same set of predictors and recompute the two sets of probabilities from the previous question. Does it make a difference to use this type of model?